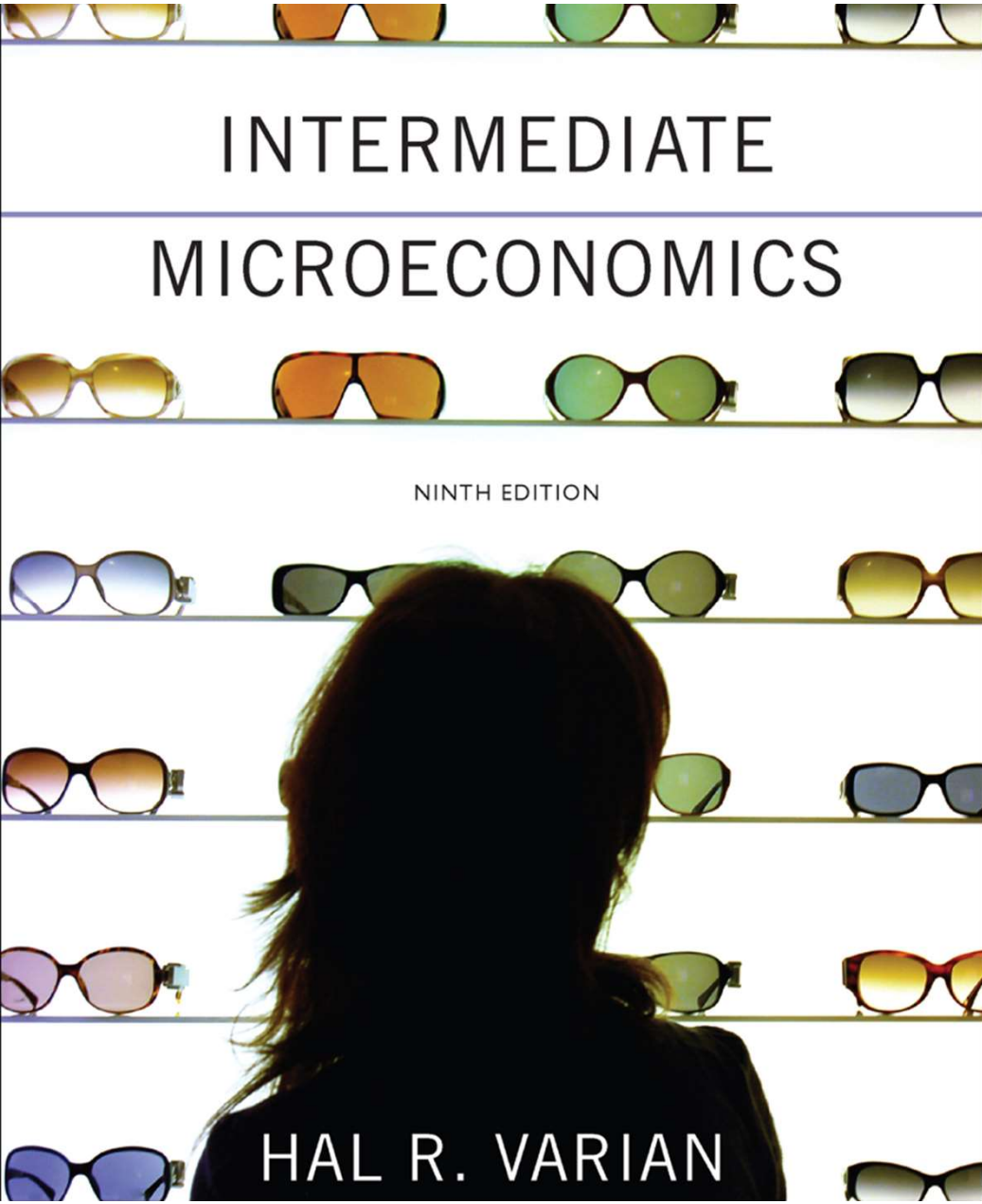


The book cover features a white background with horizontal shelves holding various pairs of sunglasses in different colors (orange, green, blue, red, purple). The title 'INTERMEDIATE MICROECONOMICS' is printed in large, black, sans-serif capital letters. Below the title, 'NINTH EDITION' is written in smaller, black, sans-serif capital letters. At the bottom, the author's name 'HAL R. VARIAN' is printed in large, black, sans-serif capital letters. A silhouette of a person's head and shoulders is visible in the lower-left foreground, looking up at the shelves.

INTERMEDIATE MICROECONOMICS

NINTH EDITION

HAL R. VARIAN

CHAPTER 15

Market Demand and Elasticity

From Individual to Market Demand Functions

Think of an economy containing n consumers, denoted by $i=1, \dots, n$.

Consumer i 's ordinary demand function for commodity 1 is

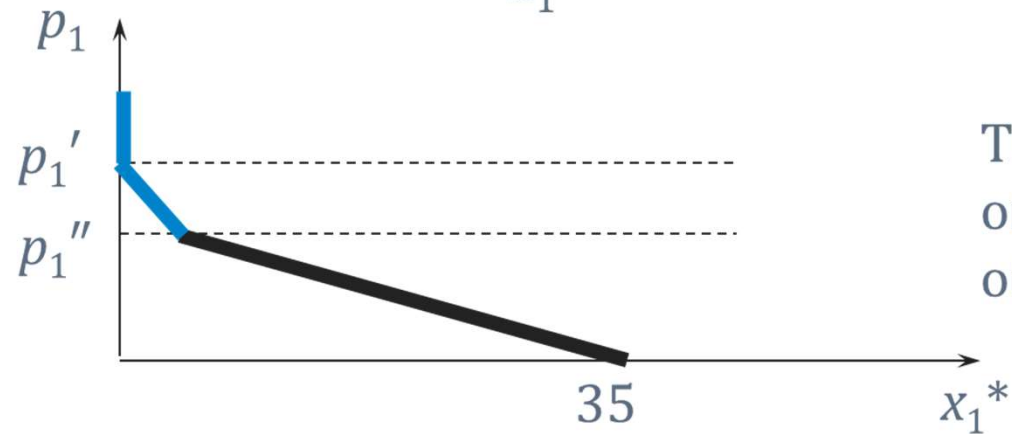
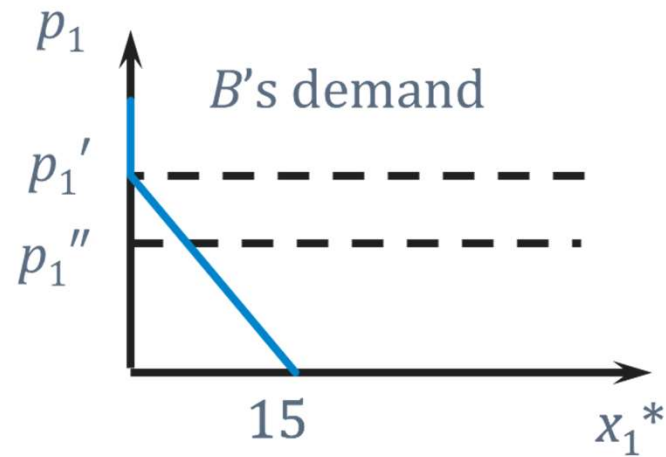
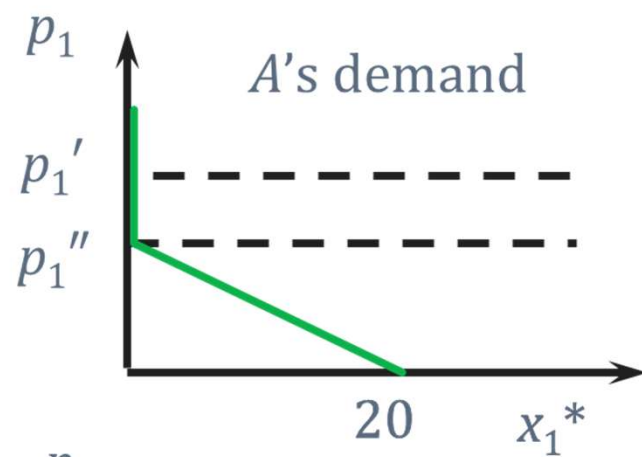
$$x_i^1(p_1, p_2, m_i).$$

When all consumers are price takers, the market demand function for commodity 1 is

$$x^1(p_1, p_2, m_1, \dots, m_n) = \sum_{i=1}^n x_i^1(p_1, p_2, m_i) \quad .$$

If all consumers are identical then $x^1(p_1, p_2, M) = nx_i^1(p_1, p_2, m)$ where $M = nm$.

From Individual to Market Demand Functions



The horizontal sum
of the demand curves
of individuals A and B

Elasticities

Elasticity measures the “sensitivity” of one variable with respect to another.

The elasticity of variable x with respect to variable y is

$$\varepsilon_{x,y} = \frac{\text{Percentage change in } x}{\text{Percentage change in } y} = \frac{\Delta x/x}{\Delta y/y} = \frac{\% \Delta x}{\% \Delta y}$$

Types of Elasticities

1. own-price elasticity of demand (change in demand x_1 with respect to change in p_1)
2. cross-price elasticity of demand (change in demand x_1 with respect to change in p_2)
3. income elasticity of demand (change in demand x_1 with respect to change in m)
4. own-price elasticity of supply (change in supply x_1 with respect to change in p_1)
5. cross-price elasticity of supply (change in supply x_1 with respect to change in p_2)

Why we need elasticity?

Why not just use the slope of a demand curve to measure the sensitivity of quantity demanded to a change in a commodity's own price?

- Because the value of sensitivity then depends upon the (arbitrary) units of measurement used for quantity demanded.

$$\text{Slope} = \frac{\Delta x_1}{\Delta p_1}$$

The slope depends on the units of measurement.

$$\varepsilon_{x,p_1} = \frac{\Delta\%x_1}{\Delta\%p_1}$$

is a ratio of percentages and so has no units of measurement.
Hence, own-price elasticity of demand is a sensitivity measure that is independent of units of measurement.

Arc and Point Elasticities

An “average” own-price elasticity of demand for commodity i over an interval of values for p_i is an **arc elasticity**, usually computed by a midpoint formula.

Elasticity computed for a single value of p_i is a **point elasticity**.

arc elasticity

$$\begin{aligned}\varepsilon_{x,p_1} &= \frac{\Delta\%x_1}{\Delta\%p_1} \\ &= \frac{\frac{x_1(\text{new}) - x_1(\text{old})}{1/2 * (x_1(\text{new}) + x_1(\text{old}))}}{\frac{p_1(\text{new}) - p_1(\text{old})}{1/2 * (p_1(\text{new}) + p_1(\text{old}))}}\end{aligned}$$

point elasticity

$$\begin{aligned}\varepsilon_{x,p_1} &= \frac{\Delta\%x_1}{\Delta\%p_1} \\ &= \frac{\Delta x_1 / x_1}{\Delta p_1 / p_1} = \frac{p_1}{x_1} \frac{\Delta x_1}{\Delta p_1} = \frac{p_1}{x_1} \frac{dx_1}{dp_1}\end{aligned}$$

Arc elasticity

Example: Change in demand with respect to its own prices

$$x_1 = 10 - 2p_1$$

Price $p_1 = 1$, demand $x_1 = 8$

New price $p_1 = 2$, new demand $x_1 = 6$

$$\frac{\text{Percentage change in demand}}{\text{Percentage change in price}} = \frac{\frac{x_1(\text{new}) - x_1(\text{old})}{1/2 * (x_1(\text{new}) + x_1(\text{old}))}}{\frac{p_1(\text{new}) - p_1(\text{old})}{1/2 * (p_1(\text{new}) + p_1(\text{old}))}} = \frac{\frac{6 - 8}{1/2 * (6 + 8)}}{\frac{2 - 1}{1/2 * (2 + 1)}} = -0,43$$

Point elasticity

Example: Change in demand with respect to its own prices

$$x_1 = 10 - 2p_1$$

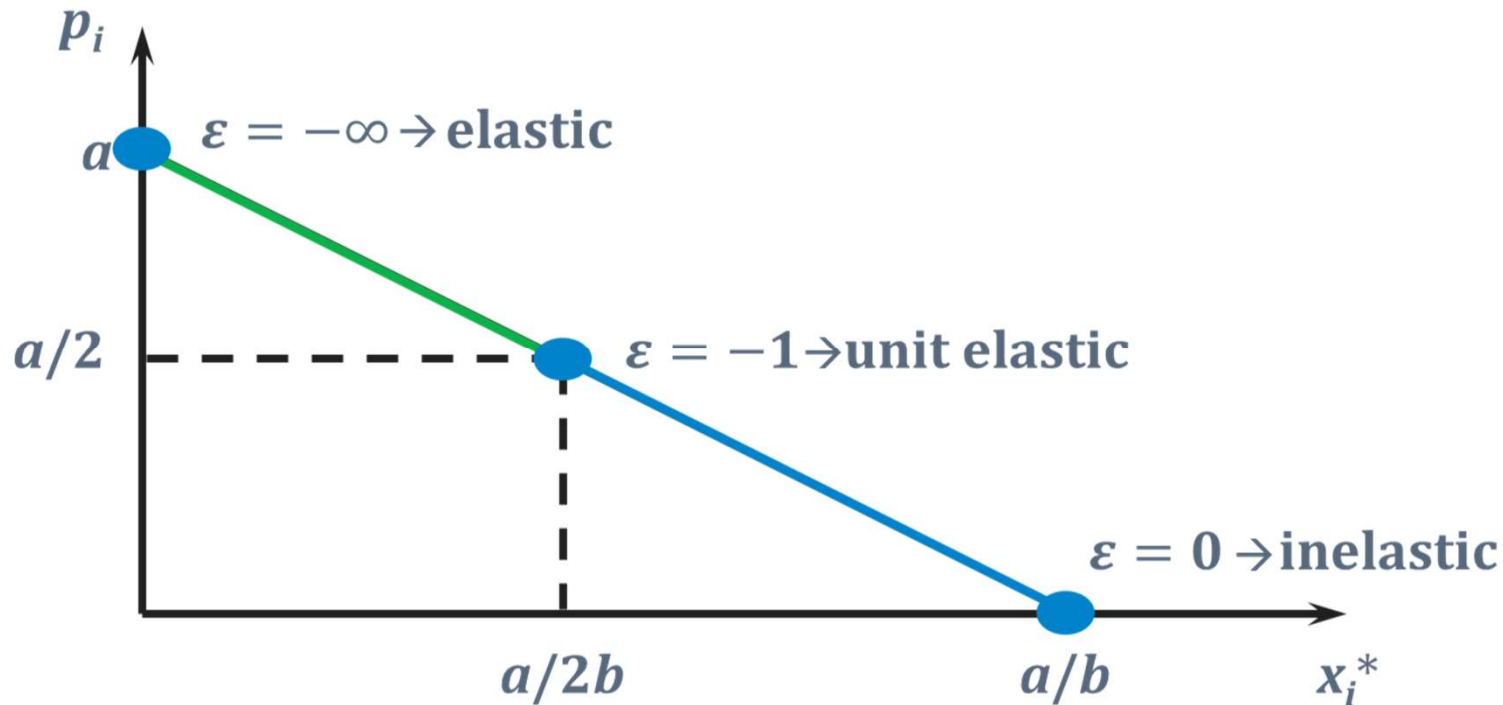
Price $p_1 = 1$, demand $x_1 = 8$

$$\frac{\text{Percentage change in demand}}{\text{Percentage change in price}} = \frac{p_1}{x_1} \frac{dx_1}{dp_1} = \frac{1}{8} (-2) = -0,25$$

Point Own-Price Elasticity

For example, suppose $p_i = a - bx_i$.
Then, $x_i = (a - p_i)/b$ and

$$\varepsilon_{x_1^*, p_1} = \frac{p_i}{(a - p_i)/b} \cdot \left(-\frac{1}{b}\right) = -\frac{p_i}{(a - p_i)}.$$



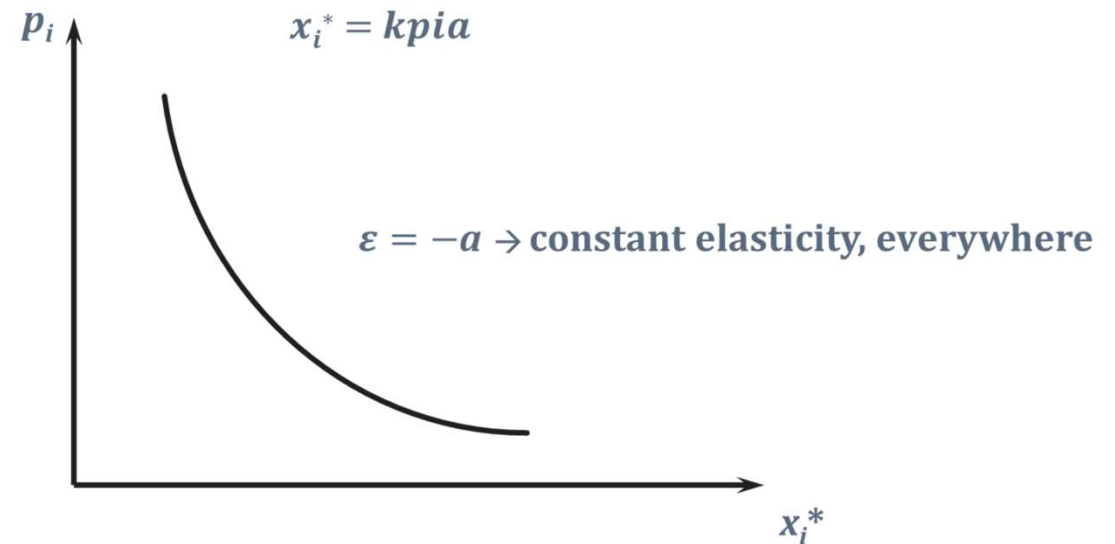
Point Own-Price Elasticity

$$\varepsilon_{x_1^*, p_1} = \frac{p_1}{x_1^*} \cdot \frac{dx_1^*}{dp_1}$$

For example, suppose $x_i^* = kp_i^a$.

$$\text{Then, } \frac{dx_1^*}{dp_1} = akp_i^{a-1}.$$

$$\text{So, } \varepsilon_{x_1^*, p_1} = -\frac{p_i}{kp_i^a} \cdot akp_i^{a-1} = -a.$$



Marginal Revenue and Own-Price Elasticity of Demand

A seller's **marginal revenue** is the rate at which revenue changes with the number of units sold by the seller.

$$MR(q) = \frac{dR(q)}{dq}$$

$p(q)$ denotes the seller's inverse demand function; i.e., the price at which the seller can sell q units. Then

$$R(q) = p(q) \cdot q.$$

$$\text{So, } MR(q) = \frac{dR(q)}{dq} = \frac{dp(q)}{dq} q + p(q) = p(q) \left[1 + \frac{q}{p(q)} \frac{dp(q)}{dq} \right]$$

$$\text{and } \varepsilon = \frac{dq}{dp} \cdot \frac{p}{q}.$$

$$\text{Therefore, } MR(q) = p(q) \left[1 + \frac{1}{\varepsilon} \right].$$

Marginal Revenue and Own-Price Elasticity of Demand

$$MR(q) = p(q) \left[1 + \frac{1}{\varepsilon} \right]$$

If $\varepsilon = -1$, then $MR(q) = 0$. (Selling one more unit does not effect revenue.)

If $-1 < \varepsilon \leq 0$, then $MR(q) < 0$. (Selling one more unit reduces revenue.)

If $\varepsilon < -1$, then $MR(q) > 0$. (Selling one more unit increases revenue.)

Marginal Revenue and Own-Price Elasticity of Demand

An example with linear inverse demand.

$$p(q) = a - bq$$

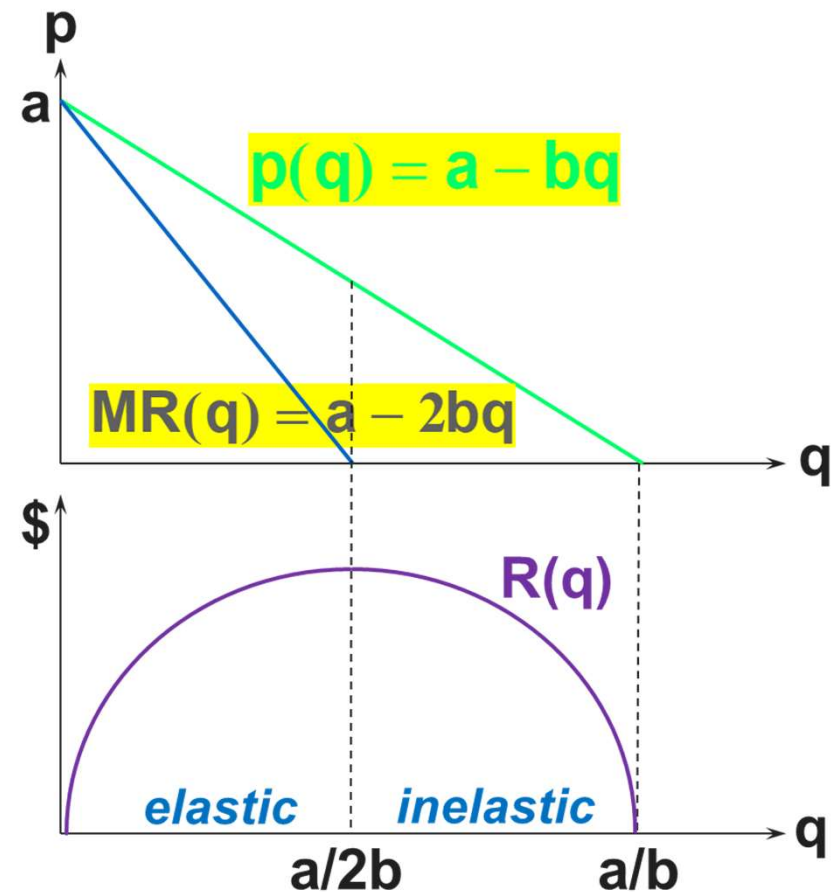
$$\text{Then, } R(q) = p(q) \cdot q = (a - bq)q.$$

$$\text{Therefore, } MR(q) = a - 2bq.$$

Demand is elastic when $0 < p < a/2b \rightarrow MR$ is positive

Demand is unit elastic when $p = a/2b \rightarrow MR = 0$

Demand is inelastic when $p > a/2b \rightarrow MR$ is negative



Real life applications

Own-Price Elasticity of Demand (USA)

Google Play apps	3.7
Cinema	2.5
Apple App Store	2
Organic milk	1.8
Air travel for non business	1.5
Vacations	0.8
Beef	0.8
Premier League football	0.7
Fruit	0.7
Air travel business	0.7
Opera	0.7
Milk	0.6
Public transport	0.4
Broadband	0.4
Wine	0.3
Theatre	0.3
Beer	0.2
Car fuel	0.2

Real life applications

Income Elasticity of Demand (USA)

Table 5
Income and price elasticities by vacation destination.

	Domestic vacations only
Income elasticities	
η_{qv} quality of travel	-0.049
η_{vq} number of vacations	0.328*
η_v total travel	0.279*
η_{qo} quality of on-site services	0.370*
η_{d_1} number of vacation days	0.050
η_D total on-site	0.420*
η_L length of vacation	-0.277*

Commodity	Income Elasticity
Beef and veal	0.4549
Pork	0.4427
Chicken	0.3645
Turkey	0.3196
Eggs	-0.0283
Cheese	0.5927
Fluid milk	-0.2209
Wheat flour	-0.1333
Rice	-0.3664
Potatoes	0.1586
Butter	0.0227
Margarine	0.1112
Apples	-0.3514
Oranges	0.4866
Bananas	-0.0429
Grapes	0.4407
Grapefruits	0.4588

Real life applications

Cross Price Elasticity of Demand (USA)

Food items	Beef	Pork	Poultry	Ground meat	Ham	Sausage	Bacon
Beef	-0.197	0.141	0.081	-0.030	0.032	-0.006	-0.020
Pork	0.099	-0.459	0.108	0.046	0.057	0.101	0.047
Poultry	0.107	0.204	-0.658	0.031	0.093	0.155	0.069
Ground meat	-0.013	0.411	0.081	-0.983	0.089	0.191	0.225
Ham	0.070	0.219	0.185	0.034	-0.639	0.072	0.059
Sausage	0.105	0.227	0.187	0.073	0.042	-0.757	0.123
Bacon	0.041	0.300	0.229	0.181	0.121	0.310	-1.183